Existing Network Details for Surface Network 1

* - Indicates pipe has been modified outside of System 1

	PN	Length (m)	Fall (m)	Slope (1:X)	I.Area (ha)	T.E. (mins)	k (mm)	HYD SECT	DIA (mm)	Section Type
	1.000	38.690	0.212	182.5	0.081	5.00	0.600	o	300	Pipe/Conduit
	1.001	6.878	0.029	237.2	0.000	0.00	0.600	o	300	Pipe/Conduit
	1.002	22.141	0.094	235.5	0.065	0.00	0.600	o	300	Pipe/Conduit
	1.003	32.563	0.137	237.7	0.091	0.00	0.600	o	300	Pipe/Conduit
	1.004	14.911	0.047	317.3	0.046	0.00	0.600	o	375	Pipe/Conduit
	1.005	30.001	0.096	312.5	0.063	0.00	0.600	o	375	Pipe/Conduit
	1.006	27.207	0.071	383.2	0.083	0.00	0.600	o	450	Pipe/Conduit
	1.007	8.613	0.022	391.5	0.010	0.00	0.600	o	450	Pipe/Conduit
	1.008	25.825	0.114	226.5	0.027	0.00	0.600	o	450	Pipe/Conduit
*	2.000	35.356	0.176	200.9	0.093	5.00	0.600	o	300	Pipe/Conduit
	2.001	20.330	0.087	233.7	0.091	0.00	0.600	o	300	Pipe/Conduit
	2.002	24.488	0.103	237.7	0.067	0.00	0.600	o	300	Pipe/Conduit
	2.003	16.640	0.071	234.4	0.047	0.00	0.600	o	300	Pipe/Conduit
	2.004	8.933	0.085	105.1	0.066	0.00	0.600	o	300	Pipe/Conduit
	1.009	24.858	0.052	478.0	0.023	0.00	0.600	o	525	Pipe/Conduit
	1.010	27.934	0.058	481.6	0.092	0.00	0.600	o	525	Pipe/Conduit
	1.011	24.973	0.045	555.0	0.192	0.00	0.600	o	600	Pipe/Conduit
	1.012	38.577	0.070	551.1	0.113	0.00	0.600	o	600	Pipe/Conduit

	PN	US/MH Name	US/CL (m)	US/IL (m)	US C.Depth (m)	DS/CL (m)	DS/IL (m)	DS C.Depth (m)	Ctrl	US/MH (mm)
	1.000	S1	93.497	91.400	1.797	93.166	91.188	1.678		1200
	1.001	S2	93.166	91.188	1.678	93.235	91.159	1.776		2100
	1.002	S3	93.235	91.159	1.776	93.356	91.065	1.991		1800
	1.003	S4	93.356	91.065	1.991	92.738	90.928	1.510		1200
	1.004	S5	92.738	90.853	1.510	92.844	90.806	1.663		1350
	1.005	S6	92.844	90.806	1.663	93.069	90.710	1.984		1200
	1.006	S7	93.069	90.635	1.984	92.807	90.564	1.793		1200
	1.007	S8	92.807	90.564	1.793	92.700	90.542	1.708		1800
	1.008	S9	92.700	90.542	1.708	92.785	90.428	1.907		1800
*	2.000	S10	93.167	91.100	1.767	93.415	90.924	2.191		2100
	2.001	S11	93.415	90.924	2.191	93.295	90.837	2.158		1200
	2.002	S12	93.295	90.837	2.158	92.994	90.734	1.960		1200
	2.003	S13	92.994	90.734	1.960	92.882	90.663	1.919		1200
	2.004	S14	92.882	90.663	1.919	92.785	90.578	1.907		1200
	1.009	S15	92.785	90.353	1.907	92.473	90.301	1.647		2100
	1.010	S16	92.473	90.301	1.647	91.932	90.243	1.164		1500
	1.011	S17	91.932	90.168	1.164	92.100	90.123	1.377		1800
	1.012	S18	92.100	90.123	1.377	92.482	90.053	1.829		1800

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Existing Network Details for Surface Network 1

PN	Length (m)	Fall (m)	Slope (1:X)	I.Area (ha)	T.E. (mins)	k (mm)	HYD SECT	DIA (mm)	Section Type
1.013	32.887	0.060	548.1	0.081	0.00	0.600	o	600	Pipe/Conduit
1.014	13.858	0.024	577.4	0.077	0.00	0.600	o	675	Pipe/Conduit
3.000	37.032	0.164	225.8	0.123	5.00	0.600	o	300	Pipe/Conduit
3.001	36.649	0.117	313.2	0.198	0.00	0.600	o	375	Pipe/Conduit
3.002	12.139	0.031	391.6	0.038	0.00	0.600	o	450	Pipe/Conduit
3.003	44.310	0.107	414.1	0.207	0.00	0.600	o	450	Pipe/Conduit
3.004	23.755	0.062	383.1	0.049	0.00	0.600	o	450	Pipe/Conduit
1.015	11.891	0.024	495.5	0.023	0.00	0.600	o	675	Pipe/Conduit
4.000	12.391	-0.034	-364.4	0.160	4.00	0.600	o	300	Pipe/Conduit
4.001	36.052	0.089	405.1	0.044	0.00	0.600	o	300	Pipe/Conduit
4.002	14.313	0.045	318.1	0.000	0.00	0.600	o	375	Pipe/Conduit
* 1.016	174.017	0.177	983.1	0.000	0.00	0.600	o	1500	Pipe/Conduit
1.017	4.362	0.043	101.5	0.000	0.00	0.600	o	1200	Pipe/Conduit
1.018	6.799	0.050	136.0	0.000	0.00	0.600	o	225	Pipe/Conduit

PN	US/MH Name	US/CL (m)	US/IL (m)	US C.Depth (m)	DS/CL (m)	DS/IL (m)	DS C.Depth (m)	Ctrl	US/MH (mm)
1.013	S19	92.482	90.053	1.829	92.327	89.993	1.734		1500
1.014	S20	92.327	89.918	1.734	92.112	89.894	1.543		1500
3.000	S21	92.650	90.750	1.600	92.671	90.586	1.785		1800
3.001	S22	92.671	90.511	1.785	92.180	90.394	1.411		1200
3.002	S23	92.180	90.319	1.411	92.262	90.288	1.524		1200
3.003	S24	92.262	90.288	1.524	92.273	90.181	1.642		1350
3.004	S25	92.273	90.181	1.642	92.112	90.119	1.543		1350
1.015	S26	92.112	89.894	1.543	92.200	89.870	1.655		2100
4.000	S27	92.110	90.400	1.410	92.166	90.434	1.432		1350
4.001	S28	92.166	90.370	1.496	92.156	90.281	1.575		1200
4.002	S29	92.156	90.215	1.566	92.200	90.170	1.655		1350
* 1.016	S30	92.200	89.870	0.830	91.300	89.693	0.107		2550
1.017	S31	91.300	89.693	0.407	91.300	89.650	0.450		1500
1.018	S32	91.300	89.650	1.425	90.700	89.600	0.875	Hydro-Brake®	3000

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Free Flowing Outfall Details for Surface Network 1

Outfall Pipe Number	Outfall Name	C. Level (m)	I. Level (m)	Min I. Level (m)	D,L (mm)	W (mm)
1.018	S33	90.700	89.600	0.000	1350	0

Simulation Criteria for Surface Network 1

Volumetric Runoff Coeff	0.750	Additional Flow - % of Total Flow	0.000
Areal Reduction Factor	1.000	MADD Factor * 10m³/ha Storage	0.000
Hot Start (mins)	0	Inlet Coeffiecient	0.800
Hot Start Level (mm)	0	Flow per Person per Day (l/per/day)	0.000
Manhole Headloss Coeff (Global)	0.500	Run Time (mins)	60
Foul Sewage per hectare (l/s)	0.000	Output Interval (mins)	1

Number of Input Hydrographs	0	Number of Storage Structures	1
Number of Online Controls	1	Number of Time/Area Diagrams	0
Number of Offline Controls	0	Number of Real Time Controls	0

Synthetic Rainfall Details

Rainfall Model	FSR	Profile Type	Summer
Return Period (years)	2	Cv (Summer)	0.750
Region	England and Wales	Cv (Winter)	0.840
M5-60 (mm)	20.000	Storm Duration (mins)	30
Ratio R	0.400		

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Online Controls for Surface Network 1

Hydro-Brake® Optimum Manhole: S32, DS/PN: 1.018, Volume (m³): 14.1

Unit Reference	MD-SHE-0136-9600-1450-9600
Design Head (m)	1.450
Design Flow (l/s)	9.6
Flush-Flo™	Calculated
Objective	Minimise upstream storage
Application	Surface
Sump Available	Yes
Diameter (mm)	136
Invert Level (m)	89.650
Minimum Outlet Pipe Diameter (mm)	150
Suggested Manhole Diameter (mm)	1200

Control Points	**Head (m)**	**Flow (l/s)**
Design Point (Calculated)	1.450	9.6
Flush-Flo™	0.427	9.6
Kick-Flo®	0.905	7.7
Mean Flow over Head Range	-	8.4

The hydrological calculations have been based on the Head/Discharge relationship for the Hydro-Brake® Optimum as specified. Should another type of control device other than a Hydro-Brake Optimum® be utilised then these storage routing calculations will be invalidated

Depth (m)	**Flow (l/s)**	**Depth (m)**	**Flow (l/s)**	**Depth (m)**	**Flow (l/s)**	**Depth (m)**	**Flow (l/s)**
0.100	4.9	1.200	8.8	3.000	13.5	7.000	20.3
0.200	8.7	1.400	9.4	3.500	14.6	7.500	21.0
0.300	9.4	1.600	10.1	4.000	15.5	8.000	21.6
0.400	9.6	1.800	10.6	4.500	16.4	8.500	22.3
0.500	9.6	2.000	11.2	5.000	17.3	9.000	22.9
0.600	9.4	2.200	11.7	5.500	18.1	9.500	23.5
0.800	8.7	2.400	12.2	6.000	18.8		
1.000	8.1	2.600	12.6	6.500	19.6		

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Storage Structures for Surface Network 1

Tank or Pond Manhole: S31, DS/PN: 1.017

Invert Level (m) 89.693

Depth (m)	Area (m²)	Depth (m)	Area (m²)	Depth (m)	Area (m²)	Depth (m)	Area (m²)
0.000	354.0	1.400	1840.0	2.800	0.0	4.200	0.0
0.200	566.0	1.600	0.0	3.000	0.0	4.400	0.0
0.400	778.0	1.800	0.0	3.200	0.0	4.600	0.0
0.600	991.0	2.000	0.0	3.400	0.0	4.800	0.0
0.800	1203.0	2.200	0.0	3.600	0.0	5.000	0.0
1.000	1416.0	2.400	0.0	3.800	0.0		
1.200	1628.0	2.600	0.0	4.000	0.0		

Manhole Headloss for Surface Network 1

PN	US/MH Name	US/MH Headloss
1.000	S1	0.500
1.001	S2	0.500
1.002	S3	0.500
1.003	S4	0.500
1.004	S5	0.500
1.005	S6	0.500
1.006	S7	0.500
1.007	S8	0.500
1.008	S9	0.900
2.000	S10	0.500
2.001	S11	1.200
2.002	S12	0.500
2.003	S13	0.500
2.004	S14	0.500
1.009	S15	0.500
1.010	S16	0.500
1.011	S17	0.500
1.012	S18	0.500
1.013	S19	0.500
1.014	S20	0.500
3.000	S21	0.500
3.001	S22	0.500
3.002	S23	0.500
3.003	S24	0.500
3.004	S25	0.500
1.015	S26	0.500
4.000	S27	0.500
4.001	S28	0.500
4.002	S29	0.500
1.016	S30	0.900
1.017	S31	0.500
1.018	S32	0.500

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Summary of Critical Results by Maximum Level (Rank 1) for Surface Network 1

Simulation Criteria

Areal Reduction Factor	1.000	Additional Flow - % of Total Flow	0.000
Hot Start (mins)	0	MADD Factor * 10m³/ha Storage	0.000
Hot Start Level (mm)	0	Inlet Coeffiecient	0.800
Manhole Headloss Coeff (Global)	0.500	Flow per Person per Day (l/per/day)	0.000
Foul Sewage per hectare (l/s)	0.000		

Number of Input Hydrographs	0	Number of Storage Structures	1
Number of Online Controls	1	Number of Time/Area Diagrams	0
Number of Offline Controls	0	Number of Real Time Controls	0

Synthetic Rainfall Details

Rainfall Model	FSR	Ratio R	0.400
Region	England and Wales	Cv (Summer)	0.750
M5-60 (mm)	20.000	Cv (Winter)	0.840

Margin for Flood Risk Warning (mm)	300.0
Analysis Timestep	2.5 Second Increment (Extended)
DTS Status	ON
DVD Status	ON
Inertia Status	ON

Profile(s)	Summer and Winter
Duration(s) (mins)	15, 30, 60, 120, 180, 240, 360, 480, 600, 720, 960, 1440, 2160, 2880, 4320, 5760, 7200, 8640, 10080
Return Period(s) (years)	100
Climate Change (%)	40

PN	US/MH Name	Storm	Return Period	Climate Change	First (X) Surcharge	First (Y) Flood	First (Z) Overflow	Overflow Act.
1.000	S1	15 Winter	100	+40%	100/15 Summer			
1.001	S2	15 Winter	100	+40%	100/15 Summer			
1.002	S3	15 Winter	100	+40%	100/15 Summer			
1.003	S4	15 Winter	100	+40%	100/15 Summer			
1.004	S5	15 Winter	100	+40%	100/15 Summer			
1.005	S6	15 Winter	100	+40%	100/15 Summer			
1.006	S7	15 Winter	100	+40%	100/15 Summer			
1.007	S8	15 Winter	100	+40%	100/15 Summer			
1.008	S9	15 Winter	100	+40%	100/15 Summer			
2.000	S10	15 Winter	100	+40%	100/15 Summer	100/15 Winter		
2.001	S11	15 Winter	100	+40%	100/15 Summer			
2.002	S12	15 Winter	100	+40%	100/15 Summer			
2.003	S13	15 Winter	100	+40%	100/15 Summer			
2.004	S14	15 Winter	100	+40%	100/15 Summer			
1.009	S15	15 Winter	100	+40%	100/15 Summer			
1.010	S16	15 Winter	100	+40%	100/15 Summer			
1.011	S17	15 Winter	100	+40%	100/15 Summer			
1.012	S18	15 Winter	100	+40%	100/15 Summer			

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Summary of Critical Results by Maximum Level (Rank 1) for Surface Network 1

PN	US/MH Name	Water Level (m)	Surcharged Depth (m)	Flooded Volume (m³)	Flow / Cap.	Overflow (l/s)	Pipe Flow (l/s)	Status	Level Exceeded
1.000	S1	93.031	1.331	0.000	0.52		39.3	SURCHARGED	
1.001	S2	92.934	1.446	0.000	0.72		37.4	FLOOD RISK	
1.002	S3	92.891	1.432	0.000	0.92		58.4	SURCHARGED	
1.003	S4	92.810	1.445	0.000	1.24		81.1	SURCHARGED	
1.004	S5	92.595	1.367	0.000	1.12		97.3	FLOOD RISK	
1.005	S6	92.542	1.361	0.000	1.27		126.8	SURCHARGED	
1.006	S7	92.401	1.316	0.000	1.20		166.9	SURCHARGED	
1.007	S8	92.315	1.301	0.000	1.69		160.7	SURCHARGED	
1.008	S9	92.269	1.277	0.000	1.03		167.4	SURCHARGED	
2.000	S10	93.168	1.768	0.743	0.64		46.0	FLOOD	1
2.001	S11	93.144	1.920	0.000	1.33		72.8	FLOOD RISK	
2.002	S12	92.976	1.839	0.000	1.62		103.4	SURCHARGED	
2.003	S13	92.741	1.707	0.000	2.01		123.5	FLOOD RISK	
2.004	S14	92.487	1.524	0.000	2.12		154.7	SURCHARGED	
1.009	S15	92.155	1.277	0.000	1.77		317.2	SURCHARGED	
1.010	S16	91.983	1.157	0.000	1.90		344.6	SURCHARGED	
1.011	S17	91.767	0.999	0.000	1.87		405.9	FLOOD RISK	
1.012	S18	91.593	0.870	0.000	1.79		441.1	SURCHARGED	

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Summary of Critical Results by Maximum Level (Rank 1) for Surface Network 1

PN	US/MH Name	Storm	Return Period	Climate Change	First (X) Surcharge	First (Y) Flood	First (Z) Overflow	Overflow Act.	Water Level (m)
1.013	S19	15 Winter	100	+40%	100/15 Summer				91.359
1.014	S20	15 Winter	100	+40%	100/15 Summer				91.130
3.000	S21	15 Winter	100	+40%	100/15 Summer				92.394
3.001	S22	15 Winter	100	+40%	100/15 Summer				92.240
3.002	S23	15 Winter	100	+40%	100/15 Summer				91.897
3.003	S24	15 Winter	100	+40%	100/15 Summer				91.782
3.004	S25	15 Winter	100	+40%	100/15 Summer				91.291
1.015	S26	15 Winter	100	+40%	100/15 Summer				90.981
4.000	S27	15 Winter	100	+40%	100/15 Summer				91.421
4.001	S28	15 Winter	100	+40%	100/15 Summer				91.245
4.002	S29	720 Winter	100	+40%	100/15 Summer				90.920
1.016	S30	720 Winter	100	+40%					90.919
1.017	S31	720 Winter	100	+40%	100/600 Winter				90.909
1.018	S32	720 Winter	100	+40%	100/15 Summer				90.926

PN	US/MH Name	Surcharged Depth (m)	Flooded Volume (m³)	Flow / Cap.	Overflow (l/s)	Pipe Flow (l/s)	Status	Level Exceeded
1.013	S19	0.706	0.000	1.91		461.8	SURCHARGED	
1.014	S20	0.537	0.000	2.61		483.2	SURCHARGED	
3.000	S21	1.344	0.000	1.02		69.3	FLOOD RISK	
3.001	S22	1.354	0.000	1.70		172.3	SURCHARGED	
3.002	S23	1.128	0.000	1.84		190.6	FLOOD RISK	
3.003	S24	1.044	0.000	2.10		297.6	SURCHARGED	
3.004	S25	0.660	0.000	2.33		318.3	SURCHARGED	
1.015	S26	0.412	0.000	3.83		791.9	SURCHARGED	
4.000	S27	0.721	0.000	4.11		102.7	SURCHARGED	
4.001	S28	0.575	0.000	2.51		127.0	SURCHARGED	
4.002	S29	0.330	0.000	0.11		9.3	SURCHARGED	
1.016	S30	-0.451	0.000	0.05		103.5	OK	
1.017	S31	0.016	0.000	0.02		29.9	SURCHARGED	
1.018	S32	1.051	0.000	0.30		9.6	SURCHARGED	

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